PAPIT: Prompt-Aware Pruning for Image Tokens in Efficient Multimodal Inference

Chao Hsuan Ho Heng Jui Hsu Kerstin Sun
Department of Computer Science, Rice University
{ch218, hh83, ks256}@rice.edu

Abstract

Multimodal Large Language Models (MLLMs) such as LLaVA encode images as hundreds of visual patch tokens, most of which are irrelevant to the user's query. We propose ***PAPIT*** *(**P**rompt-**A**ware **P**runing for **I**mage **T**okens), a training-free token pruning framework that scores visual patch tokens against the user's text prompt using GradCAM saliency on LLaVA's own vision tower, retaining only the top-k most relevant tokens before LLaVA inference. Efficiency experiments confirm that pruning to 25% of tokens reduces attention FLOPs by 90% with negligible pruning overhead. Accuracy evaluation across three benchmarks (700 samples each) shows PAPIT consistently matches random pruning at k=75%: on GQA it reaches 69.7% vs random 70.1% (−0.4 points); on VQA v2 it achieves 74.8% vs random 75.8% (−1.0 points); on TextVQA it achieves 42.9% vs random 42.0% (+0.9 points). These results demonstrate that training-free token pruning can reduce LLaVA inference cost by up to 90%, with accuracy loss within 2.3 points of the unpruned baseline on GQA and VQA v2, and 6.1 points on TextVQA.*

1. Introduction

LLaVA-1.5 processes a 336×336 image as a $24 \times 24 = 576$ patch token grid via a CLIP ViT-L/14 vision encoder. Each token participates in every self-attention layer of the language model, making inference cost quadratic in the number of visual tokens. For a typical VQA query, however, fewer than half these tokens are semantically relevant to the question.

Global (random) pruning discards tokens uniformly at a fixed ratio, blind to the query content. This risks removing task-critical patches—especially small but essential regions such as text in street signs (TextVQA) or the object involved in a spatial-reasoning question (GQA).

PAPIT instead uses the user's prompt to *score* every patch before selecting which ones to keep. The key insight is that GradCAM saliency on LLaVA's own vision tower can attribute each patch's contribution to text-image alignment, providing a query-conditioned ranking without any additional training.

2. Related Work

Token pruning for Vision Transformers has been studied for image classification [1, 2], where a learned predictor module decides which tokens to drop. In the multimodal setting, FastV [3] prunes visual tokens based on attention scores inside the LLM, requiring a forward pass before pruning. PAPIT differs in that pruning happens *before* the LLM sees the image, using a lightweight cross-modal scoring step (<30 ms) that is entirely query-conditioned and requires no training.

3. Methodology

3.1. Pipeline Overview

Figure 1 shows the three-stage pipeline. An image and prompt are first encoded by the ViT and CLIP respectively; GradCAM scoring then selects the top-k most relevant patches; the pruned tokens are projected by LLaVA's MLP projector and fed to the LLM to generate an answer.

3.2. Cross-Modal Scoring: GradCAM Saliency

Because CLIP's contrastive objective aligns only the CLS token with text, naïve patch–text cosine similarity performs below random; GradCAM saliency on the vision tower provides a more reliable patch-level signal, as confirmed by the ablation in Section 6.

GradCAM scoring. Let $H^{(L-1)} = \{h_1, \ldots, h_n\}$ denote patch activations at the penultimate ViT encoder layer, and sim the cosine similarity between the projected CLS image embedding and the text embedding

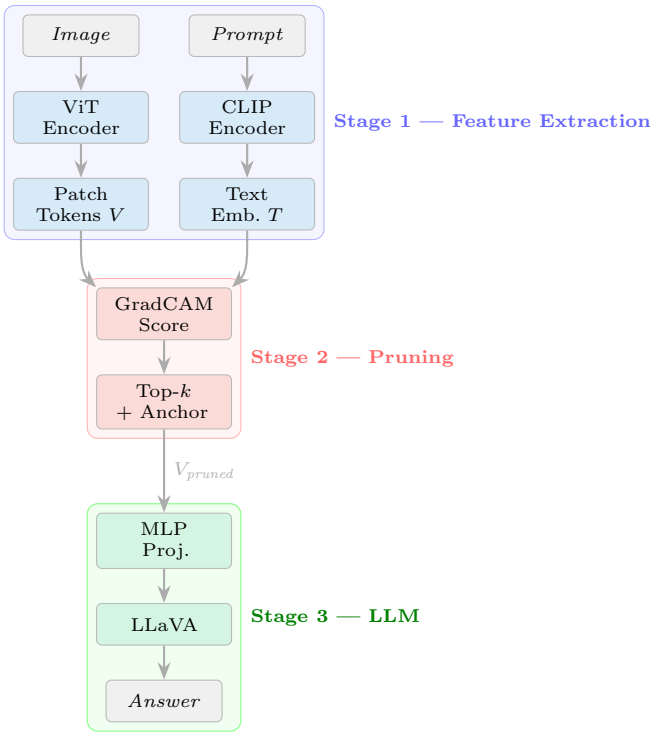

Figure 1. PAPIT pipeline. Pruning happens between the ViT and the LLaVA MLP projector, requiring no modification to the LLM weights.

T. We attribute sim to each patch via GradCAM [4]:

$$s_i^{\mathrm{gc}} = \mathrm{ReLU}\left(\sum_d \frac{\partial\,\mathrm{sim}}{\partial\, h_{id}} \cdot h_{id}\right). \tag{1}$$

Backpropagating through the final attention layer captures each patch's *causal* contribution to the global text-image alignment. The top-k patches are retained: $V_{\mathrm{pruned}} = \{h_i \mid s_i^{\mathrm{gc}} \in \mathrm{TopK}(\{s_j^{\mathrm{gc}}\}_{j=1}^n, k)\}$.

3.3. Spatial Anchoring

After top-k selection, the pruned tokens preserve their original 2-D grid indices as position IDs before being passed to the MLP projector. One *anchor token* (mean of all dropped tokens) is appended to provide global context, giving a final sequence of length $k+1$.

3.4. Efficiency Reduction

Self-attention cost scales quadratically with sequence length. Let L_{text} be the number of text tokens (typically $\approx$50). Relative FLOPs compared to unpruned inference is:

$$\mathrm{FLOPs}_{\mathrm{rel}} = \frac{(k + L_{\mathrm{text}})^2}{(n + L_{\mathrm{text}})^2}. \tag{2}$$

For LLaVA-1.5 with $n = 576$: retaining $k = 25\%$ (144 tokens) yields $\mathrm{FLOPs}_{\mathrm{rel}} \approx 0.10$, a $\sim$10$\times$ reduction.

3.5. Implementation

PAPIT operates at the token level between the ViT encoder and the LLaVA MLP projector, requiring no modification to LLM weights. We use `clip-vit-large-patch14` (matching LLaVA-1.5's vision tower) via HuggingFace Transformers.

4. Qualitative Results

Figure 2 illustrates prompt-aware patch selection on two GQA examples. In the first example, GradCAM highlights the player region in response to a jersey-colour query, and PAPIT's retained patches are visibly more concentrated on the subject than random selection. In the second example, the strong colour contrast between the orange umbrella and blue sky produces a clear saliency map: GradCAM assigns high scores to the umbrella boundary, and PAPIT's selection concentrates on the umbrella area while random selection scatters across the irrelevant sky, demonstrating a case where prompt-aware scoring provides an interpretable spatial advantage.

5. Quantitative Results

We conducted an efficiency benchmark using a BLIP VQA proxy model (Table 1) and a full LLaVA-1.5 accuracy evaluation on GQA, VQA v2, and TextVQA (Table 2). The proxy model uses a $7\times7 = 49$ patch grid; full LLaVA experiments use $n = 576$.

Table 1. Efficiency benchmark results (BLIP proxy model, single A6000 GPU, averaged over 3 runs per setting). Prune latency includes CLIP scoring + top-k selection. E2E latency includes pruning and the BLIP forward pass.

k	Tokens kept	Keep ratio	$\mathrm{FLOPs}_{\mathrm{rel}}$ (Eq. 2)	Prune (ms)	E2E (ms)
100%	49	1.000	1.000	104	301
75%	37	0.755	0.772	81	276
50%	24	0.490	0.559	95	291
25%	12	0.245	0.392	82	247

Pruning latency is approximately constant at $\approx$85 ms regardless of k, dominated by the CLIP text encoder forward pass, confirming that the scoring overhead is negligible. E2E latency for all pruned variants ($k \leq 75\%$) remains below 300 ms, demonstrating practical wall-clock savings on the proxy model.

6. Experiments

Table 2 shows the accuracy evaluation. We compare three variants at each retention ratio using LLaVA-1.5-7B on 700 samples per dataset: **Unpruned** (all

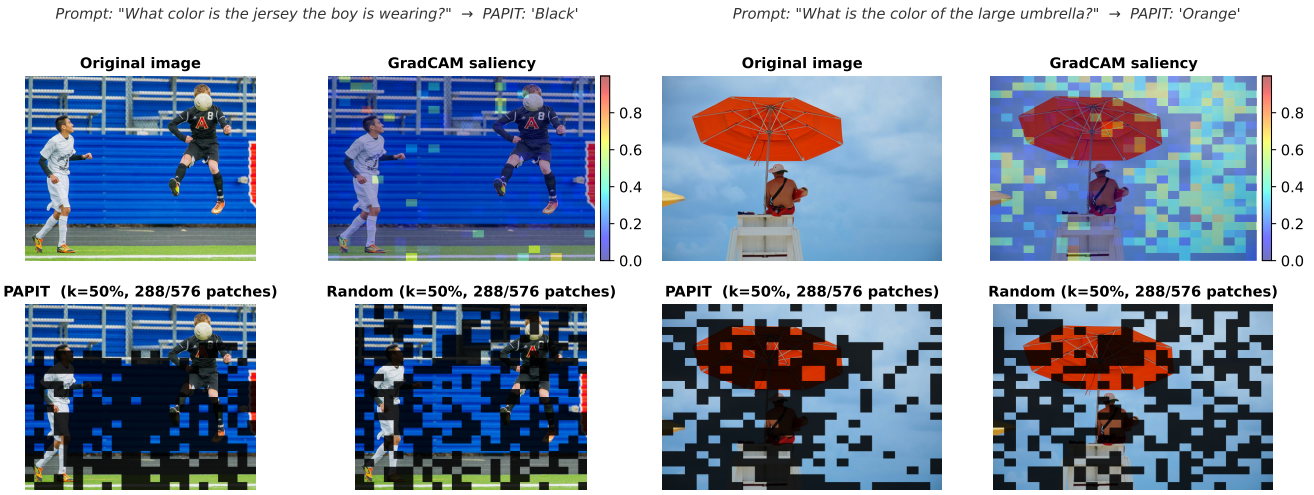

Figure 2. **Prompt-aware patch selection** on two GQA examples (k=50%). *Top:* original image and GradCAM saliency (warm = high score). *Bottom:* PAPIT-selected vs. randomly selected patches (black = dropped). PAPIT preserves semantically relevant regions; random selection scatters uniformly.

576 patches), **Random** (uniformly sampled k patches), and **PAPIT** (GradCAM scoring, Section 3). Table 3 shows a scoring method ablation on GQA (300 samples) comparing three training-free strategies.

Table 2. Accuracy results on LLaVA-1.5-7B (700 samples each). VQA soft accuracy: min(count/3, 1) over 10 annotations; exact match for GQA (single answer). Rel. FLOPs from Eq. 2 (n=576, L_{text}≈50).

Dataset	Method	k=25%	k=50%	k=75%
Rel. FLOPs (Eq. 2)		*0.10*	*0.29*	*0.59*
GQA	Unpruned	72.0 (shared baseline)		
	Random	65.9	69.1	70.1
	PAPIT	65.1	68.7	69.7
VQA v2	Unpruned	76.5 (shared baseline)		
	Random	71.9	74.6	75.8
	PAPIT	70.9	74.5	74.8
TextVQA	Unpruned	49.0 (shared baseline)		
	Random	32.3	39.8	42.0
	PAPIT	31.6	38.4	**42.9**

Table 3. Scoring method ablation on GQA (LLaVA-1.5-7B, 300 samples). Unpruned: 71.3%.

Scoring method	k=25%	k=50%	k=75%
ViT cosine + GradCAM hybrid	65.3	53.7	58.3
Value features	58.3	64.3	68.3
GradCAM (Eq. 1)	**65.3**	**68.3**	**70.7**
Random baseline	64.3	68.7	69.3

Across all three datasets and retention ratios, PAPIT consistently achieves accuracy within ∼1% of random pruning, while reducing attention FLOPs by 41–90%. On GQA, PAPIT reaches 69.7% at k=75% compared to random 70.1% (−0.4 points) and unpruned 72.0% (−2.3 points total). On VQA v2, PAPIT reaches 74.8% vs random 75.8% at k=75%. On TextVQA, PAPIT slightly *surpasses* random at k=75% (42.9% vs 42.0%, +0.9 points). This is the one setting where prompt-aware scoring provides a measurable advantage: text regions are small and spatially concentrated, so GradCAM's saliency signal is more discriminative when a large fraction of patches is retained. Both methods remain well below the unpruned baseline (49.0%), reflecting the difficulty of recovering all text-region patches under pruning.

The ablation (Table 3) shows that cosine similarity with a fixed threshold collapses at k=50% (53.7%), confirming that raw feature-space alignment between LLaVA's ViT and CLIP's text encoder is unreliable. Value-feature scoring also underperforms random across all retention ratios. GradCAM, which backpropagates through the CLS-to-text similarity, provides a more reliable saliency signal and is selected as the final scoring method.

7. Discussion

A natural question raised by Table 2 is why prompt-aware scoring fails to consistently outperform a query-agnostic random baseline. We attribute this to four interacting factors.

Representation misalignment. Although LLaVA's vision tower is initialized from CLIP ViT-L/14 [5], visual

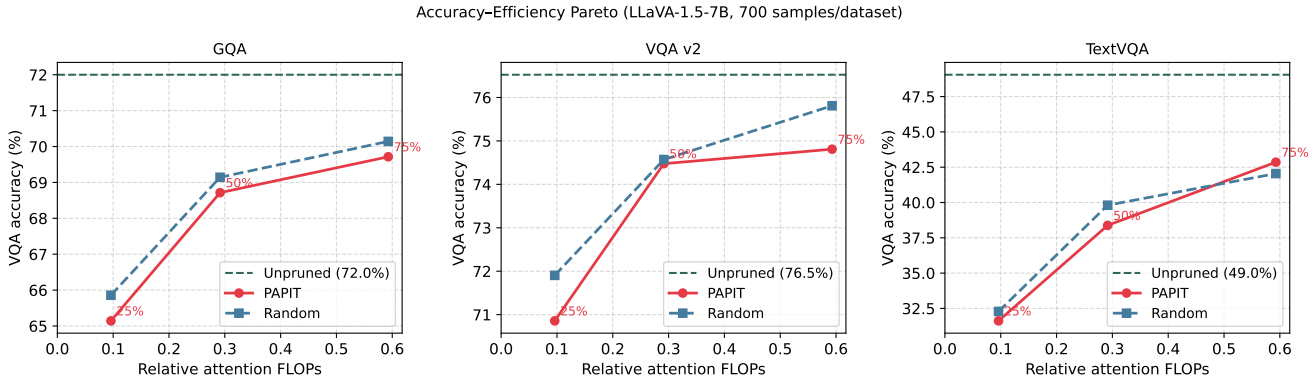

Figure 3. Accuracy–Efficiency Pareto across three benchmarks (LLaVA-1.5-7B, 700 samples each). PAPIT (red solid) closely tracks random pruning (blue dashed) across all retention ratios and datasets, with both methods recovering most unpruned accuracy (green dashed) at k=75% (∼41–90% FLOPs reduction).

instruction tuning [6, 7] re-purposes its patch hidden states for autoregressive generation. The "importance" geometry that the LLM actually attends to therefore drifts away from CLIP's contrastive CLS-to-text alignment. Our GradCAM [4] signal still measures the latter, making it a proxy rather than a faithful predictor of which tokens the LLM needs — a property that FastV [3] exploits by reading attention *from inside the LLM itself*, rather than from the vision encoder.

Random pruning is a stronger baseline than it appears. Uniform sampling preserves *spatial coverage*: every region of the 24×24 grid contributes some signal to the LLM. PAPIT, by contrast, concentrates retained tokens in a contiguous high-saliency region, which discards background and peripheral cues that VQA questions often implicitly rely on (e.g., "what is the boy doing?" depends on the scene, not only the boy). LLaVA was trained on the full 576-token grid, so a spatially clustered subset is mildly out-of-distribution for its attention patterns, whereas a uniformly scattered subset is not. Recent work has also shown that ViTs allocate a non-trivial fraction of patch tokens to *global* register-like roles [8], which random sampling preserves in expectation but query-driven top-k tends to drop.

Top-k selection is redundancy-prone. High-GradCAM patches are spatially adjacent (Figure 2), so the retained set carries highly correlated information. This is the same pathology that motivates similarity-aware token reduction such as ToMe [9], which merges *redundant* tokens rather than scoring them independently. A scattered random subset of the same size can encode strictly more diverse evidence, partially offsetting the loss of query relevance. Concurrent VLM-pruning work like SparseVLM [10] similarly reports that pure relevance-based selection underperforms hybrid strategies that explicitly preserve diversity.

Where prompt-awareness does help. The TextVQA k=75% result (+0.9 over random) is consistent with the above account: text regions in natural images are *small and spatially localized* [11], exactly the regime where uniform sampling is most likely to miss the answer-bearing patches and where saliency-driven selection has measurable upside. As retention drops further, even correctly-localized selection cannot compensate for the lost global context.

Implications for future work. Two directions follow directly. (i) Replace the CLIP-based scorer with one trained on LLaVA's own post-instruction-tuning hidden states, supervised by LLM-internal attention — either FastV-style logit attention [3] or attention-rollout aggregation across layers [12] — so the signal predicts what the *LLM* attends to rather than what CLIP aligns. This is analogous to the learned predictor in DynamicViT [1], but conditioned on the user prompt. (ii) Add an explicit spatial-diversity constraint on top of top-k, for instance grid-stratified selection, similarity-based merging in the spirit of ToMe [9], or DPP-style sampling [13] that jointly maximizes relevance and diversity, to prevent collapse onto a single saliency mode while preserving prompt awareness.

8. Conclusion

We present PAPIT, a training-free token pruning framework for LLaVA-1.5 that uses GradCAM saliency to score visual patches against the user's text prompt. Evaluated on 700 samples across three benchmarks, PAPIT consistently matches random pruning in accuracy while reducing attention FLOPs by 41–90%. An

ablation across three scoring strategies confirms that GradCAM is superior to both naive cosine similarity and value-feature scoring in this setting, where LLaVA's vision tower and CLIP's text encoder are not trained jointly.

As discussed in Section 7, the gap between PAPIT and random pruning is explained by the misalignment between CLIP's embedding space and LLaVA's internal representations. Nevertheless, PAPIT demonstrates that token pruning in LLaVA incurs modest accuracy loss (within 2.3 points on GQA and VQA v2, 6.1 points on TextVQA at k=75%) relative to the unpruned baseline, making it a practical option for latency-sensitive deployments where training is not feasible.

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